MAKING SENSE OF INFINITY AMONG MATHEMATICS STUDENT TEACHERS IN MALAYSIA INSTITUTES OF TEACHER EDUCATION

NG LEE FONG

UMS
UNIVERSITI MALAYSIA SABAH

FACULTY OF PSYCHOLOGY AND EDUCATION
UNIVERSITI MALAYSIA SABAH
2018

UNIVERSITI MALAYSIA SABAH

BORANG PENGESAHAN STATUS TESIS

JUDUL : **MAKING SENSE OF INFINITY AMONG MATHEMATICS STUDENT TEACHERS IN MALAYSIA INSTITUTES OF TEACHER EDUCATION**

IZAJAH : **DOKTOR FALSAFAH (PENILAIAN DALAM PENDIDIKAN)**

Saya **NG LEE FONG**, sesi **2011-2018**, mengaku membenarkan tesis Doktoral ini disimpan di perpustakaan Universiti Malaysia Sabah dengan syarat-syarat kegunaan seperti berikut:-

1. Tesis ini adalah hak milik Universiti Malaysia Sabah.
2. Perpustakaan Universiti Malaysia Sabah dibenarkan membuat salinan untuk tujuan pengajian sahaja.
3. Perpustakaan dibenarkan membuat salinan tesis ini sebagai bahan pertukaran antara institut pengajian tinggi.
4. Sila tandakan (/):

☐ SULIT (Mengandungi maklumat yang berdarjah keselamatan atau kepentingan Malaysia seperti yang termaktub di dalam AKTA RAHSIA 1972)

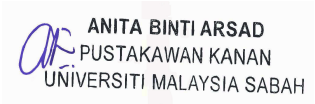
☐ TERHAD (Mengandungi maklumat TERHAD yang telah ditentukan oleh organisasi/badan di mana penyelidikan dijalankan)

☐ TIDAK TERHAD

UMS
UNIVERSITI MALAYSIA SABAH

Disahkan Oleh,





NG LEE FONG
PT20119066

Tarikh: 06 Ogos 2018

(Professor Dr. Vincent Pang Ah Fook)
Penyelia



(Dr. Chin Kin Eng @ Sporty Chin)
Penyelia Bersama

MAKING SENSE OF INFINITY AMONG MATHEMATICS STUDENT TEACHERS IN MALAYSIA INSTITUTES OF TEACHER EDUCATION

NG LEE FONG

UMS
UNIVERSITI MALAYSIA SABAH

THESIS SUBMITTED IN FULFILLMENT FOR THE DEGREE OF DOCTOR OF PHILOSOPHY

FACULTY OF PSYCHOLOGY AND EDUCATION
UNIVERSITI MALAYSIA SABAH
2018

DECLARATION

I, Ng Lee Fong, hereby declare that this thesis "Making Sense of Infinity among Mathematics Student Teachers in Malaysia Institutes of Teacher Education" is an original work done by me for the award of the degree of Doctor of Philosophy in the Faculty of Psychology and Education. I also declare that the materials in this thesis are of my own except quotations, excerpts, equations, summaries and references, which have been duly acknowledged.



06 August 2018

NG LEE FONG
PT20119066

UMS
UNIVERSITI MALAYSIA SABAH

CERTIFICATION

NAME : **NG LEE FONG**

MATRIC NO : **PT20119066**

TITTLE OF THESIS : **MAKING SENSE OF INFINITY AMONG MATHEMATICS STUDENT TEACHERS IN MALAYSIA INSTITUTES OF TEACHER EDUCATION**

DEGREE : **DOCTOR OF PHILOSOPHY (EVALUATION IN EDUCATION)**

DATE OF VIVA : **11 JULY 2018**

CERTIFIED BY;

Signature

1. **MAIN SUPERVISOR**
   Professor Dr. Vincent Pang

2. **CO-SUPERVISOR**
   Dr. Chin Kin Eng @ Sporty Chin

UNIVERSITI MALAYSIA SABAH

ACKNOWLEDGEMENT

I am pleased to take this opportunity to express my gratitude to my supervisor, Professor Vincent Pang and co-supervisor, Dr Chin Kin Eng for all their support and advice which they had given me during this study and the encouragement that brought the thesis to completion. I find myself very fortunate working with both of them, each having their own expertise and view-points. Although occasionally there were differences in ideas leaving me confused I am very grateful for all the knowledge and wisdom they have taught me. I am particularly indebted to Professor Vincent Pang who has painstakingly read and re-read the drafts, commenting on them and correcting my grammar.

I would like to thank Dr. Azlena Zainal, Chang Siew Lee and Tan Mei Chin who helped to correct my English in the early version. I would like to thank my colleagues at the Research and Innovation Department at selected Institution of Teacher Education Malaysia who helped to administer the questionnaire and collect the data on behalf of me. My thanks also go to all student-teachers who volunteered to participate in the survey and interview. I am also greatly indebted to PM Dr. Ho Chong Mun, PM Dr. Aini Janteng, Dr. Gan Wei Ling, Dr. Chiam Sun May, Dr Kok Boon Shiong for all of their help in developing my instrument. To all my HLP friends, you have offered me unlimited encouragement and have been my support in all areas. Thank you for sharing this crazy graduate school journey with me.

Special thanks go to my supportive husband, Wong Ken Keong, who encouraged, pushed, praised and intimately comforted me throughout the journey. His unwavering support, tireless help and endless patience made it all possible. Thank you more than I can say. To my siblings, Kyle Wong Junkai and Eily Wong Yili, both of you have demonstrated rare and amazing patience throughout my lengthy working sessions over the last four years. Thank you, handsome boy and pretty girl, for motivating me to keep reaching for excellence.

NG LEE FONG

06 August 2018

UNIVERSITI MALAYSIA SABAH

ABSTRACT

This thesis attempts to explain mathematics student teachers' varying concepts of infinity based on the ways of making sense. Two pilot studies, carried out at Institutes of Teacher Education (IPG) located in Sabah and Sarawak, tested possible items that can lead to how mathematics student teachers make sense of infinity. The theoretical framework in the research focuses on mathematical thinking, which describes the ways of making sense about infinity based on perception, operation and reason. Although the research focuses on infinity, the theory from the framework is applicable to other topics in the field of mathematics. Using the experience gained through the pilot studies, the main research was carried out at 13 selected IPGs, which involved 238 student teachers majoring in mathematics. There are four of the student teacher took part in the interview sessions. Three types of concepts in infinite divisibility thinking were discussed in this study, namely potential infinity thinking, actual infinity thinking and problematic thinking. Mostly student teachers were making sense about their conceptions of infinity by perception and a few of them are able to think at the higher stage by operation and reasoning. The findings indicate that the sense-making affected student teachers' thinking about infinity concept. The inferential analysis results show that there is a significant relation between cardinality construct with conception of infinity but has no significant relationships between equalities construct and limit construct with the conception of infinity. The Kruskal Wallis test analysis indicated that there are significant differences among the infinite divisibility thinking types in the conception of infinity. The last idea is about supportive conceptions and problematic conceptions in making sense of mathematics. Supportive conception is noticeable among the minority of the student teachers. That is not the whole story of the research. This research revealed that a supportive conception might contain problematic aspects and a problematic conception might contain supportive aspects. The research found there are problematic conceptions in all constructs of infinity among the mathematics student teachers and their knowledge related to infinity should be explored further.

UNIVERSITI MALAYSIA SABAH

ABSTRAK

MEMBINA PEMAHAMAN INFINITI DALAM KALANGAN GURU PELATIH MATEMATIK DI INSTITUT PENDIDIKAN GURU MALAYSIA

Tesis ini cuba menerangkan konsep infiniti yang berbeza dalam kalangan guru pelatih opsyen matamatik berdasarkan cara mereka membina pemahaman (making sense). Dua kajian rintis telah dijalankan di Institut Pendidikan Guru (IPG) yang terletak di Sabah dan Sarawak, bagi menguji item-item yang mungkin dapat menbantu guru pelatih matematik membina pemahaman sebenar infiniti. Kerangka teori dalam kajian ini berfokuskan kepada pemikiran dalam bidang matematik yang menerangkan cara membina pemahaman berkaitan infiniti berdasarkan persepsi (perception), operasi (operation) dan penaakulan (reasoning). Walaupun kajian ini tertumpu kepada topik infiniti, teori dari kerangka kajian ini dapat diterapkan kepada topik yang lain dalam bidang matematik. Berdasarkan pengalaman yang dilaksanakan dalam kajian rintis, kajian sebenar telah dijalankan di 13 buah IPG terpilih dengan melibatkan 238 orang guru pelatih dari pengkhususan matematik. Empat orang guru pelatih telah mengambil bahagian dalam sesi temu bual. Terdapat tiga jenis konsep pemikiran pembahagian infiniti yang dibincangkan dalam kajian ini, iaitu pemikiran infiniti potensi (potential infinity thinking), pemikiran infiniti sebenar (actual infinity thinking) dan pemikiran bermasalah (problematic thinking). Kebanyakan guru pelatih membina kefahaman tentang konsep infiniti berdasarkan persepsi dan hanya sebilangan di antara mereka yang dapat berfikir pada tahap yang lebih tinggi berdasarkan operasi dan penaakulan. Hasil kajian ini menunjukkan bahawa cara mereka membina pemahaman mempengaruhi pemikiran guru pelatih tentang konsep infiniti. Analisis inferens menunjukkan bahawa terdapat hubungan yang signifikan antara konstruk kardinaliti (cardinality construct) dengan konsepsi infiniti tetapi tidak ada hubungan yang signifikan antara konstruk kesamaan (equilities consruct) dan konstruk had (limit construct) dengan konsepsi infiniti. Analisis ujian Kruskal Wallis menunjukkan bahawa terdapat perbezaan yang signifikan di antara jenis pemikiran pembahagian dalam konsep infiniti. Idea terakhir adalah mengenai konsepsi sokongan (supportive conception) dan konsepsi bermasalah (problematic conception) dalam membina pemahaman matematik. Konsepsi sokongan dapat dikesan pada sebahagian kecil guru pelatih. Itu bukan gambaran keseluruhan dalam kajian ini. Kajian ini mendedahkan bahawa konsepsi sokongan mungkin mengandungi aspek yang bermasalah dan konsepsi bermasalah juga mungkin mengandungi aspek-aspek sokongan. Dapatan kajian menunjukkan bahawa terdapat konsep-konsep yang bermasalah dalam semua konstruk infiniti di kalangan guru pelatih dan pengetahuan mereka yang berkaitan dengan infiniti harus diterokai dengan lebih lanjut.

UNIVERSITI MALAYSIA SABAH

TABLE OF CONTENTS

		Page
TITLE		i
DECLARATION		ii
CERTIFICATION		iii
ACKNOWLEDGEMENT		iv
ABSTRACT		v
ABSTRAK		vi
TABLE OF CONTENTS		vii
LIST OF TABLES		xi
LIST OF FIGURES		xiv
LIST OF ABBREVIATIONS		xv
LIST OF APPENDICES		xvi
CHAPTER 1: INTRODUCTION		1
1.1	Introduction	1
1.2	Background of the Research	3
1.3	Statement of the Problem	5
1.4	Research Objectives	9
1.5	Research Questions	9
1.6	Hypotheses	10
1.7	Significance of the Research	10
	1.7.1 Student Teachers	11
	1.7.2 Teachers and IPG Lecturers	11
	1.7.3 Curriculum Designers	12
1.8	Definitions	12
1.9	Limitations of the Research	16
1.10	Organisation of the Thesis	18
1.11	Conclusion	18

UMS
UNIVERSITI MALAYSIA SABAH

CHAPTER 2: REVIEW OF LITERATURE 20
2.1 Introduction 20
2.2 Theories of Mathematical Thinking 21
2.2.1 The Compression of Learning in Mathematics 27
2.2.2 The Blending of Theories to Making Sense in Mathematical Thinking 33
2.3 The Conceptions in Mathematical Learning 40
2.4 Mathematical Thinking as Defined in Malaysian Mathematics Curriculum 42
2.5 Studies about Infinity 45
2.5.1 The History of Infinity 46
2.5.2 The Intuition of Infinity 49
2.5.3 The Conception of Infinity Thinking 51
2.5.4 The Divisibility Thinking 53
2.5.5 The Cardinality 55
2.5.6 The Equality 58
2.5.7 The Limit in Infinity 60
2.6 Supportive Conceptions and Problematic Conceptions in Changing Contexts of Infinity 62
2.7 Recent Studies Related to Infinity 65
2.8 Theoretical Framework 67
2.8.1 The Concept Image: Pre-conception of Infinity 68
2.8.2 Making Sense in Mathematical Thinking 69
2.9 The Conceptual Framework 71
2.10 Conclusion 72

CHAPTER 3: METHODOLOGY 73
3.1 Introduction 73
3.2 Research Paradigm, Strategies and Procedure 74
3.3 Research Location, Population and Sample 81
3.4 Instrumentation 83
3.4.1 Pilot 1 Study: The Development of ITA Instrument 85
3.4.2 Pilot 2 Study: The Reliability and Validity of Instrument 104
3.4.3 Interrater Reliability 116

UMS
UNIVERSITI MALAYSIA SABAH

	3.4.4 The Follow-up Interview	122
3.5	Data Analysis	129
	3.5.1 Quantitative Data Analysis	129
	3.5.2 Qualitative Data Analysis	133
3.6	Ethical Issues	135
3.7	Conclusion	136
CHAPTER 4: FINDINGS		137
4.1	Introduction	137
4.2	Demography of Respondents	138
4.3	The Divisibility	140
	4.3.1 The Types of Infinity Thinking	140
	4.3.2 The Ways of Sense Making based on Types of Infinity Thinking	141
4.4	The Infinity Constructs	149
	4.4.1 Cardinality Construct Performance and the Ways of Sense Making	150
	4.4.2 Equality Construct Performance and the Ways of Sense Making	161
	4.4.3 Limit Construct Performance and the Ways of Sense Making	168
4.5	Normality of Data	172
4.6	The Correlations between Constructs and Conception of Infinity	173
	4.6.1 The Correlations between Finite and Infinite Cardinality with the Conception of Infinity	174
	4.6.2 The Correlations between Equality and the Conception of Infinity	174
	4.6.3 The Correlations between Limit and the Conception of Infinity	175
4.7	The Difference in the Conceptions based on Divisibility Thinking	175
4.8	The Ways of Making Sense about the Conception of Infinity	176
4.9	Supportive and Problematic Conceptions in Infinity	190
	4.9.1 The Cardinality in Infinity	190
	4.9.2 The Equality in Infinity	194

UNIVERSITI MALAYSIA SABAH

4.9.3 The Limit in Infinity 198
4.9.4 The Conception on Defining the Infinity Term 200
4.9.5 The Conception on Infinity based on Operation 202
4.9.6 Defining Infinity 209
4.10 Conclusion 211

CHAPTER 5: CONCLUSION AND RECOMMENDATIONS 212
5.1 Introduction 212
5.2 The Summary of Research 212
5.3 The Discussion of Findings 214
5.3.1 The Divisibility 215
5.3.2 The Infinity Constructs' Performance and Ways of Sense Making 218
5.3.3 The Correlations between Constructs and Conception of Infinity 221
5.3.4 The Difference in the Conceptions based on Infinite Divisibility 222
5.3.5 The Ways of Sense Making in Conception of Infinity 223
5.3.6 The Supportive and Problematic Conceptions in Infinity 224
5.4 Implications of the Findings 227
5.4.1 Implications to the Student teachers 228
5.4.2 Implications to the Trained Teachers and IPG Lecturers 228
5.4.3 Implications to the Administrators 229
5.4.4 Implications to the Curriculum Designers 230
5.4.5 Implications on Development of Knowledge 231
5.5 Suggestions for Further Research 232
5.6 Conclusion 233

REFERENCES 235

UMS
UNIVERSITI MALAYSIA SABAH

LIST OF TABLES

		Page
Table 2.1:	Global Stages of Cognitive Development	22
Table 2.2:	Local Cycle of Cognitive Development	31
Table 2.3:	Links Between the Proposed Theoretical Frameworks	37
Table 2.4:	Comparison of Objectives Related to Mathematical Thinking in Malaysian School Mathematics Curricula	44
Table 3.1:	The Population of Mathematics Programme Student Teachers	82
Table 3.2:	The Sample based on Intake of Respondents	83
Table 3.3:	The Pilot 1 ITA Questionnaire	87
Table 3.4:	Justifications on Pilot 1 ITA	89
Table 3.5:	The Construct of ITA Instrument	90
Table 3.6:	Ways of Making Sense in Item 1a	92
Table 3.7:	Ways of Making Sense in Item 1b	93
Table 3.8:	Ways of Making Sense in Item 1c	94
Table 3.9:	Responses in Item 3a	95
Table 3.10:	Responses in Item 3b	97
Table 3.11:	Responses for Equality in Item 4a	97
Table 3.12:	Responses for Equality in Item 4b	98
Table 3.13:	The Result of Question 5a and 5b	99
Table 3.14:	The Result of Question 10a	100
Table 3.15:	The Result of Question 10b	101
Table 3.16:	The Result of Question 10c	101
Table 3.17:	The Level of Confident in Answering the Questionnaire	102
Table 3.18:	Result on Second Pilot Study	105
Table 3.19:	The Coefficient of Confident in Reliability	106
Table 3.20:	The Summary Rasch Analysis for Person Reliability	108
Table 3.21:	The Summary Rasch Analysis for Item Reliability	109
Table 3.22:	The ITA's Content Validity Analysis Rated by Experts	113
Table 3.23:	Correlation on Construct Validity of ITA Instrument	115
Table 3.24:	Item Fit Table	116
Table 3.25:	Calculation of Percent Agreement on Question 1	119

UMS
UNIVERSITI MALAYSIA SABAH

Table 3.26:	Calculation of Percent Agreement on Question 3	120
Table 3.27:	Calculation of Percent Agreement on Question 4	121
Table 3.28:	Calculation of Percent Agreement on Question 10	122
Table 3.29:	Data Analysis Table	130
Table 3.30:	The Rule of Thumb for a Correlation Coefficient	132
Table 3.31:	Examples of Codes	135
Table 4.1:	Demography of the Respondents	139
Table 4.2:	Types of Infinity Thinking	140
Table 4.3:	Explanations and Sense Making for Item 1a	142
Table 4.4:	Explanations and Sense Making for Item 1b	144
Table 4.5:	Explanations and Sense Making for Item 1c	146
Table 4.6:	Student Teacher's Performance by Items in the Constructs	150
Table 4.7:	Descriptive Statistics of the Student Teachers' Performance on Cardinality Construct	151
Table 4.8:	Descriptive Statistics of the Student Teachers' Performance based on Finite Cardinality and Infinite Cardinality	151
Table 4.9:	Explanations and Sense Making for item 2a	152
Table 4.10:	Explanations and Sense Making for item 2b	153
Table 4.11:	Explanations and Sense Making for item 2c	154
Table 4.12:	Explanations and Sense Making for item 2d	155
Table 4.13:	Explanations and Sense Making for item 2e	157
Table 4.14:	Explanations and Sense Making for item 2f	159
Table 4.15:	Explanations and Sense Making for item 2g	160
Table 4.16:	Descriptive Statistics of the Student Teachers' Performance on Equality Construct	161
Table 4.17:	Explanations and Sense Making for Item 3a	162
Table 4.18:	Explanations and Sense Making for Item 3b	164
Table 4.19:	Explanations and Sense Making for Item 3c	166
Table 4.20:	Explanations and Sense Making for Item 3d	167
Table 4.21:	Descriptive Statistics of the Student Teachers' Performance on Limit Construct	168
Table 4.22:	Explanations and Sense Making for Item 4a	170
Table 4.23:	Explanations and Sense Making for Item 4b	171
Table 4.24:	Tests of Normality	173

UNIVERSITI MALAYSIA SABAH

Table 4.25:	The Correlation between Finite Cardinality and Infinite Cardinality with the Conception of Infinity	174
Table 4.26:	The Correlations between Equality and the Conception of Infinity	175
Table 4.27:	The Correlations between Limit and the Conception of Infinity	175
Table 4.28:	The Kruskal-Wallis's Result by Mean Rank	176
Table 4.29:	The Kruskal-Wallis's Result by Chi-Square	176
Table 4.30:	Descriptive Statistics of the Student Teachers' Performance on Conception of Infinity	177
Table 4.31:	Explanations and Sense Making for Item 5a	178
Table 4.32:	Explanations and Sense Making for Sub-item 5b1 (a + ∞)	180
Table 4.33:	Explanations and Sense Making for Sub-item 5b2 (a - ∞)	181
Table 4.34:	Explanations and Sense Making for Sub-item 5b3 (a x ∞)	183
Table 4.35:	Explanations and Sense Making for Sub-item 5b4 (a / ∞)	184
Table 4.36:	Explanations and Sense Making for Sub-item 5b5 (∞ + ∞)	186
Table 4.37:	Explanations and Sense Making for Sub-item 5b6 (∞ - ∞)	187
Table 4.38:	Explanations and Sense Making for Sub-item 5b7 (∞ x ∞)	188
Table 4.39:	Explanations and Sense Making for Sub-item 5b8 (∞ / ∞)	189

UMS
UNIVERSITI MALAYSIA SABAH

LIST OF FIGURES

		Page
Figure 2.1:	Spectrum of Outcomes from Increasing Comparison of Symbolism	29
Figure 2.2:	Outline Cognitive Development based on Tall's Hypothesis	32
Figure 2.3:	Actions and Objects in the Building of Various Mathematical Knowledge Structures to Mathematical Thinking Development	35
Figure 2.4:	The Projection of Points from Smaller Circle to Bigger Circle	58
Figure 2.5:	Research Theoretical Framework	68
Figure 2.6:	The Research Conceptual Framework	72
Figure 3.1:	Research Plan	76
Figure 3.2:	Three Stages of Research	77
Figure 3.3:	Research Procedure	80
Figure 3.4:	The Question 1 of Pilot 1 ITA	91
Figure 3.5:	The Question 3 of Pilot 1 ITA	94
Figure 3.6:	The Question 4 of Pilot 1 ITA	97
Figure 3.7:	The Question 5 of Pilot 1 ITA	99
Figure 3.8:	The Question 10 of Pilot 1 ITA	100
Figure 3.9:	Mean values of level of confidence in responding to the mathematics items	103
Figure 3.10:	The Application of Process Model in Deductive Category	133
Figure 4.1:	The Answer for Item 1a from Respondent 151	148
Figure 4.2:	The Answer for Item 1b from Respondent 151	148
Figure 4.3:	The Answer for Item 1c from Respondent 151	148
Figure 4.4:	Box Plot for Normality Test	173
Figure 4.5:	The Score based on Items in Question 5	177
Figure 4.6:	The Answer of Item 5a, Informant S4	201
Figure 4.7:	The Answer of Item 5b, Informant S1	203
Figure 4.8:	The Answer of Item 5b, Informant S2	205

UMS
UNIVERSITI MALAYSIA SABAH

LIST OF ABBREVIATIONS

MOE	**-**	Ministry of Education
IPGM	-	The Institute of Teacher Education Malaysia (*Institut Pendidikan Guru Malaysia)*
IPG	-	The Institute of Teacher Education (*Institut Pendidikan Guru)*
APOS	-	Action-Process-Object-Schema
SOLO	-	Structure of the Observed Learning Outcome
VMS	-	Visually Moderated Sequences'
ITA	-	Infinity Thinking Assessment
CVI	**-**	Content Validity Index
MNSQ	**-**	INFIT mean square
Z- stds	**-**	Z-standard score

UMS
UNIVERSITI MALAYSIA SABAH

LIST OF APPENDICES

	Page
APPENDIX A: Information Letter and Expert Detail	248
APPENDIX B: Evaluation form of Content Experts	253
APPENDIX C: Research Permission Letter from Malaysian Ministry of Education	257
APPENDIX D: Research Permission Letter from Institute of Teacher Education Malaysia (IPGM)	259
APPENDIX E: Pilot 1 ITA instrument	261
APPENDIX F: ITA Instrument	265
APPENDIX G: Example of answered ITA	276
APPENDIX H: Example of Consent Form for Interview (S2)	278
APPENDIX I: Example Part of Transcript (S2)	283
APPENDIX J: Example of Interview's Data Verification Form (S2)	286
APPENDIX K: Example of Coded Transcript (S2)	290
APPENDIX L: The Correlation between Construct (Cardinality, Equality and Limit) with the Conception of Infinity among the Student Teachers in IPG	296
APPENDIX M: Kruskal Wallis Test Result	297
APPENDIX N: The Part of Definition of Notation of Infinity by 238 Respondents	298

UNIVERSITI MALAYSIA SABAH

CHAPTER 1

INTRODUCTION

1.1 Introduction

The Institute of Teacher Education (*Institut Pendidikan Guru*, IPG) is a higher institution which trains future primary school teachers for the Ministry of Education, Malaysia. The institution runs under The Institute of Teacher Education Malaysia (*Institut Pendidikan Guru Malaysia*, IPGM). Besides training new teachers, it also comes up with various programs for teachers to upgrade themselves professionally. IPG was upgraded from a college to a university institute that offers degree courses in 2005. In line with the national education philosophy, the institute aims to produce competent and spirited educators through a dynamic teacher development program towards a world-class education. Thus, the quality of teachers should be evaluated in many different perspectives and aspects, including the way of thinking. Mathematics is a core subject in the school curriculum of Malaysia up till the secondary level. In the primary school syllabus, the curriculum is organized into four areas of learning (number and operations, measurement and geometry, relation and algebra, and statistics and probability). The Mathematics Standard Curriculum Document has structured the curriculum design by putting emphasis on producing students with mathematical thinking through four aspects, namely learning contents, attitudes and values, mathematical skills, and processes involving mathematics (Ministry of Education (MOE), 2010).

Behind calculus are the concepts of a limit, infinity and real numbers. According to Monaghan (1986), many students experience difficulty with the algebraic manipulation involved in the first calculus course but they are actually facing cognitive difficulties with the concepts of limits, infinity and in real numbers. On the other hand, little attention is paid to the notion of infinity in the mathematics subject in school. The main reason is that school mathematics is

UNIVERSITI MALAYSIA SABAH

essentially conceived as an ensemble of solving techniques. The idea of mathematics as a coherent and structurally organized body of knowledge is not systematically conveyed to the student. Teachers aim for their students to have the feeling of grandeur and appreciate the beauty of mathematics. Certainly, theorems and proofs are taught besides solving procedures but nowadays, the coherent and harmonic structure has been diluted from the system (Fischbein, Jehiam and Cohen, 1995).

The word *infinity* has often been used in the field of mathematics but many of us may not be able to provide an accurate description or definition for it. It has been proven that infinity is one of the most difficult concepts for people to understand and most mathematical studies related to the concept of infinity are about students' conception and reasoning in the calculus subject (Aztekin, Arikan and Sriraman, 2010). Chin and Tall (2012) suggested that the "teachers should be proactive to the changing needs of the student in new situations to build on previous success and realise that what worked before will need a new approach to make sense of the new situation". As a result, they consider that making sense of mathematical thinking could be through perception, operation and reason. Therefore, mathematics teachers should master the concept of infinity and know how to apply it correctly when teaching subjects with regard to infinity. However, the considerations must be given to how school students make sense of mathematical thinking.

This research looks into the conception of infinity based on how student teachers from IPG cope with their sense to make meaning of infinity. Data for the research were gathered by a questionnaire named the Infinity Thinking Assessment (ITA), which was administrated to 18 classes of student teachers intake taken from 13 IPG in Malaysia. The analyses for quantitative data and qualitative data are reported in Chapter 4 and the discussion on this research is reported in Chapter 5.

UNIVERSITI MALAYSIA SABAH

1.2 Background of the Research

The Malaysian school curriculum is committed to developing the child holistically along the physical, emotional, spiritual and intellectual dimensions, as reflected in the National Education Philosophy. It aims to develop the potential of individuals through quality education by building a generation that is capable in thinking and committed to our country. MOE is continuously reviewing the curriculum to ensure the implementation of the curriculum in schools equips students with the knowledge, skills, and values to meet present and future challenges. Starting in 2011, the primary school curriculum has been reorganized. This reorganization will take into account the on-going continuity to the next level. In the teaching and learning of mathematics, the emphasis is given to the mastery of knowledge and comprehension to enable students to apply the mathematical concepts, principles and processes learned (MOE, 2013). Emphasis is also given to the development of mathematical thinking of students through the process of teaching and learning in the classroom based on the principles of problem solving, communication, reasoning, association, and making representation and use of technology in mathematics (MOE, 2013).

Mathematics requires a way of thinking and needs a particular kind of intelligence. Intelligence varies from one individual to another due to the contrast in the language or physical ability (Tall, 2007). Mathematical thinking starts growing in a child and continues into mature adulthood. It is considered a long-term development of knowledge (Tall, 2007). Mathematical thinking involves seeing patterns in daily life, geometric reasoning, thinking logically, recognizing the relationship between numbers, having a sense of the problems, and developing strategies to solve mathematical problems (Tall, 2008). Meanwhile, concepts of infinity arise by reflecting on finite experiences and imagining to the infinite situation (Tall, 2001). There are inconsistencies by selecting a finite list of specific properties from which the conception of a formal infinity is built by formal deduction derived in the twentieth century by the formal approach to mathematics (Tall, 1992b). Different formal systems may be developed with different properties of finite numbers such as counting, ordering or arithmetic. Particularly, counting and ordering lead to the cardinal and ordinal number theory. The properties of

UNIVERSITI MALAYSIA SABAH

arithmetic lead to ordered fields that may contain infinite and infinitesimal quantities (Tall, 2001).

Therefore, this research is concerned with student teachers' infinity thinking concept in the Malaysian education system and focused on the theory of making sense of mathematical thinking and mathematics learning in the infinity concept. Infinity is considered as one of the advanced mathematical thinking concepts during the transition phase from the later years of school to college and university besides function, limits, and mathematical proof (Tall, 1992b). Schoenfeld (1992) described mathematics as a subject of life and very closely related in the minds around us and ourselves. He also stressed that motivating students to move beyond the rules would lead to a focus on finding the rules, rather than just memorizing procedures and exploring patterns, not just memorizing formulas and formulating assumptions because mathematical thinking is fundamental to mathematics education. To think mathematically does not necessarily mean that the individual is thinking in signs, symbols, and therefore can be done in any situation. Hung (2000) emphasised that the process through which students engage in the search and discovery of concept-referential meanings is critical to mathematical thinking and allows mathematics to be meaningful and useful in the real world. Besides that, mathematical thinking is important in education and learning mathematics and for teaching mathematics which is a highly complex activity (Kaye, 2007).

This research will use the term of *perception, operation,* and *reason* to continue the evolution of theories of mathematical thinking. There are other frameworks which are similar to this research; the terms *intuitive, algorithmic* and *formal* were used by Fischbein (1982), while Bruner (1971) suggested the terms *enactive*, *iconic,* and *symbolic* modes of communication. His term of symbolism included natural language and the special mathematical languages of arithmetic and logic. Personal thinking will be built up with mathematical thinking. According to Chin (2014), the personal conception of thinking might consist of supportive and problematic aspects. If individuals are able to operate using their supportive aspects successfully in any given contexts, *supportive conceptions* would be gained.

UNIVERSITI MALAYSIA SABAH

In contrast, the problematic aspects could cause *problematic conceptions* when they are sufficient to impede the progress of learning.

Infinity is an abstract concept describing something without any limit and relevant in a number of fields, predominantly mathematics and physics. In mathematics, infinity is not taught directly in one topic but inserted in a specific topic that describes unbounded values. Infinity is a meaningful mathematical concept. Individuals will accept definitions, theorems and the proof of some ideas of infinity but using its concepts in the real situation and process of interpreting are two different things (Fischbein, Tirosh, and Hess, 1979). Therefore, in mathematical thinking, there are two fundamental different aspects to consider. First, the idea is pure with the final structure that is logical and unquestionable. Second, there are some concepts which remain unsolved (complex), contradictory and related to our intuition difficulties of psychology (Fischbein, Tirosh and Hess, 1979). According to Monaghan (2001), there are at least three ways mathematicians (formally) view infinite numbers: limit ordinals, infinite cardinals, and infinite non-standard number. In addition, regarding to the question " $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots$ ", the usual answer was either infinite or 'tends to 1'. Very few students claim that the sum is 1 (Fischbein, Tirosh and Melamed, 1981). Students need divisibility thinking to get their meaning because according to Fischbein (2002), psychologically, it is really tends to 1. The difference types of views and pre-conceptions held by individuals delayed meaningful learning of infinity.

UMS
UNIVERSITI MALAYSIA SABAH

1.3 Statement of the Problem

The importance of mathematics in primary schools in Malaysia is written in their goals of developing the students' understanding and awareness of the power of mathematics in generalization and abstraction so that they can apply the knowledge and skills effectively in their daily life. When applying problem solving that involves mathematics, Kaye (2007) claimed that it is part of mathematical thinking. If students are able to think mathematically, it is an important outcome of schooling that may be used to stimulate mathematical thinking about numbers or as an introduction to algebra. When students learn mathematical symbols from

primary to higher education, they will encounter many symbols as part of the lessons. However, there is no clear definition of mathematical thinking in the mathematics curriculum of Malaysian schools. Mathematical thinking is not explicitly explained in the intended curriculum, though it is stated as one of the objectives in the secondary school mathematics curriculum. There are still very limited studies with regard to mathematical thinking in the local literatures (Lim and Yong, 2007) and specifically, there is no research about infinity thinking among student teachers in Malaysia.

In general, researchers presented their findings about students' conceptions of infinity as a personal reflection, instruction, and intuitions combined. They sought to uncover the naive interpretation of a concept that has surprised and impressed the minds for centuries. The nature of the counterintuitive and abstract infinity provides a very attractive opportunity for investigation. Moreover, infinity is a concept full of self-confidence that may stem from religion or philosophy (Mamolo, 2007). There is a contradictory nature of infinity as considered in the process of mathematical thinking (Fischbein, Tirosh, and Hess, 1979). According to Tall (2001), there are several types of concepts of infinity that may arise by reflecting on finite experiences and imagining. He affirmed that an individual has the natural and formal conception of infinity with its own coherent properties. Natural infinity is derived from personal conception by reflecting on finite experiences and the formal infinity concept is built by formal deduction from a finite list of properties. Yoop, Burroughs and Lindaman (2011) stated that it is worth to study in a large scale about the understanding in pre-service teachers with the concept of 0.999… and 1 or other related equalities. The reasons are the misunderstandings about concepts related to the concept of infinity can impact the teaching of the elementary school mathematics curriculum. For example, for the question of 0.999…999…, does it equal to one, or just less? School students must say "less", and even the teachers would say the same but in a mathematician's view, this decimal expansion equals one (Tall, 1981).

However, there is a lack of research studies that examined elementary student teachers' perceptions about infinity (Kattou, Micheal, Kontoyianni, Christou, and Philippou, 2010) and this fact has served as a motivation to conduct this